

# AUTOMATED WEEKLY NEWSLETTER GENERATOR

## Capstone Project Documentation and Deliverable Guide

### Nairobi Daily Weekly Newsletter Automation

Sample Github link- <https://github.com/markarif/Automated-Weekly-Newsletter-Generator>

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## 1. Project Overview

**The Problem** Manually checking 5 Kenyan news sites every Friday, pulling headlines and writing a newsletter by hand was taking too long. This workflow automates the entire process end to end.

### How it works

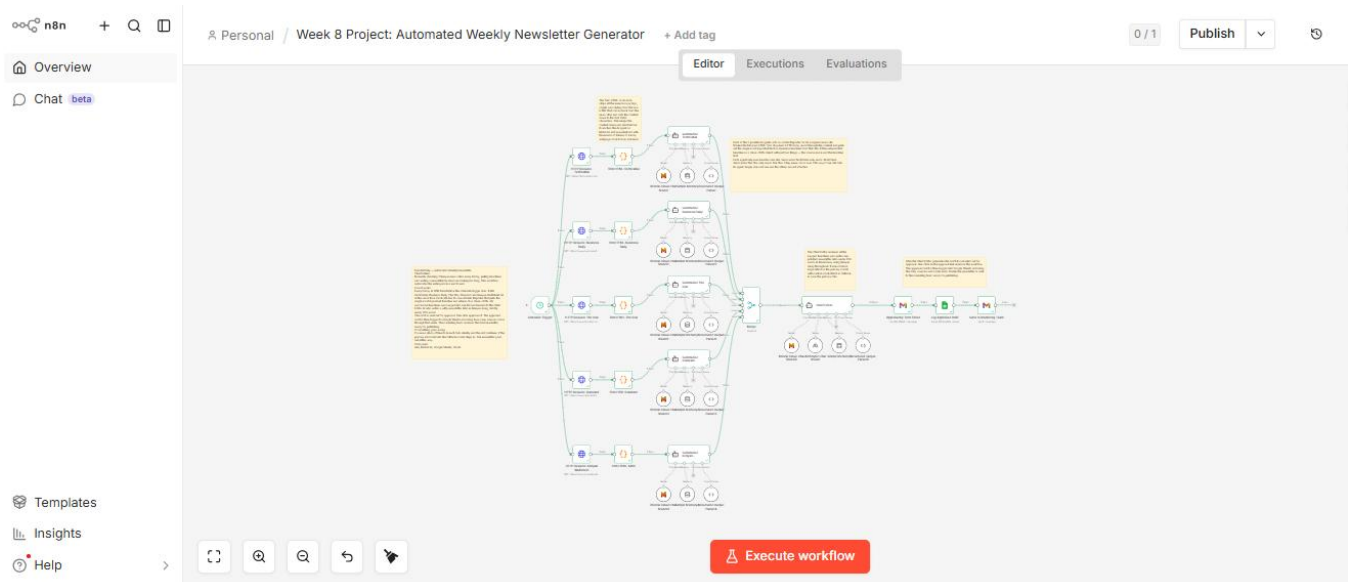
Every Friday at 2PM Nairobi time the schedule trigger fires. It hits TechCabal, Business Daily, The Star, Standard and Kenyan WallStreet all at the same time. Each site has its own AI Field Reporter that pulls the single most important headline and returns it as clean JSON. All successful headlines are merged into one list and handed to the Chief Editor AI who writes a witty newsletter intro in Kenyan slang, strictly under 200 words.

The draft is sent out for approval. One click approves it. The approved draft is then logged to Google Sheets recording how many sources came through that week. The marketing team receives the final newsletter ready for publishing.

**If something goes wrong** If a news site is offline its branch fails silently and the rest continue. If the primary AI model fails the fallback model steps in. The newsletter goes out either way.

**Tools used** n8n, Mistral AI/Claude, Google Sheets, Gmail.

## Workflow Screenshot



## **2. Workflow Explanation**

### **Step 1 - Schedule Trigger**

The Schedule Trigger is the starting point of the workflow. It runs every Friday at 2:00 PM Africa/Nairobi time. This ensures the newsletter process starts automatically every week without manual execution.

### **Step 2 - Distributed Parallel Processing**

From the trigger, the workflow splits into five branches. Each branch fetches content from one publication and immediately passes it to its own AI Field Reporter. This design improves speed and reliability because the branches run independently. Each HTTP Request node have 'Continue On Fail' enabled so a failure in one publication does not stop the entire workflow.

### **Step 3 - HTML Cleaning**

Raw website content contains scripts, styles, tags, menus, advertisements, and other noise. The Code nodes clean the HTML by removing scripts, style blocks, tags, and excess spaces before sending readable text to the AI agents.

### **Step 4 - Field Reporter AI Agents**

Each Field Reporter AI extracts the single most important tech or business headline from its assigned source. The required JSON format is {"source":"Name", "headline":"Text"}. This keeps output consistent for merging and downstream editing.

### **Step 5 - Continue On Fail**

Each HTTP Request node should have Continue On Fail enabled. This means that if one source, such as The Star, is offline or times out, that failed branch does not stop the whole workflow. The other four sources continue to report successfully.

### **Step 6 - Merge / Reduce Phase**

The Merge node receives all successful AI summaries and appends them into one combined list. This is the reduce phase because the separate branch outputs are gathered into a single editorial dataset.

### **Step 7 - Chief Editor AI**

The Chief Editor AI acts as Kamau, the witty tech editor. It receives the combined summaries and creates a short Markdown newsletter intro under 200 words using Kenyan slang such as Bazeng and Form ni gani.

### **Step 8 - Fallback Model**

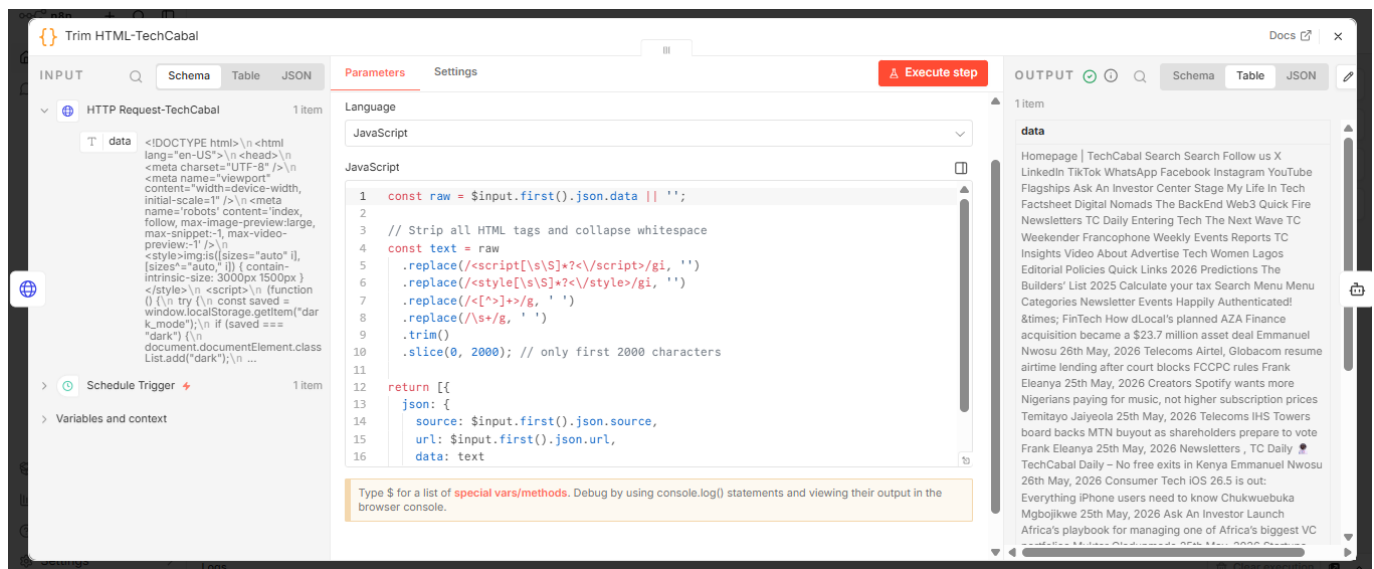
The assignment requires a backup route. If the primary Chief Editor model fails or times out, an Error Trigger or fallback branch should send the same merged summaries to a Backup AI Agent using a lighter model such as GPT-4o-mini to produce a simpler draft.

### **Step 9 - Logging and Delivery**

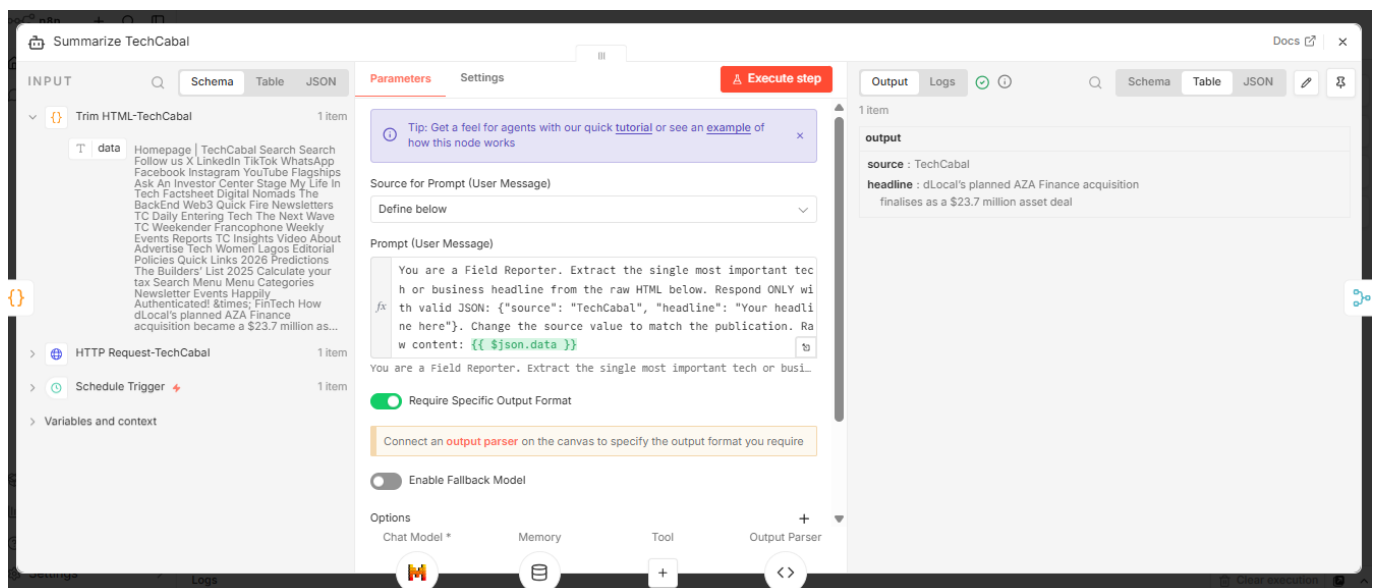
The final draft is logged in Google Sheets with the date, successful source count, draft body, and model used. Gmail then sends the Markdown draft to the marketing team for approval.

## 4. Screenshot for Deliverables

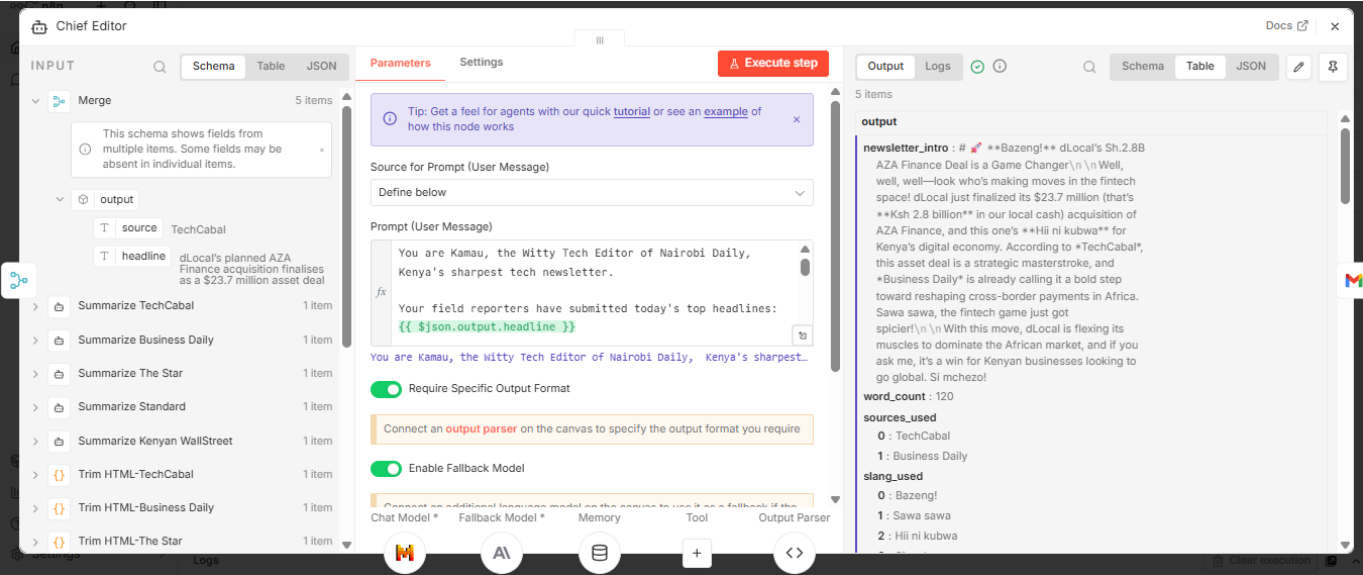
### Trim HTML — Trim HTML Code Node



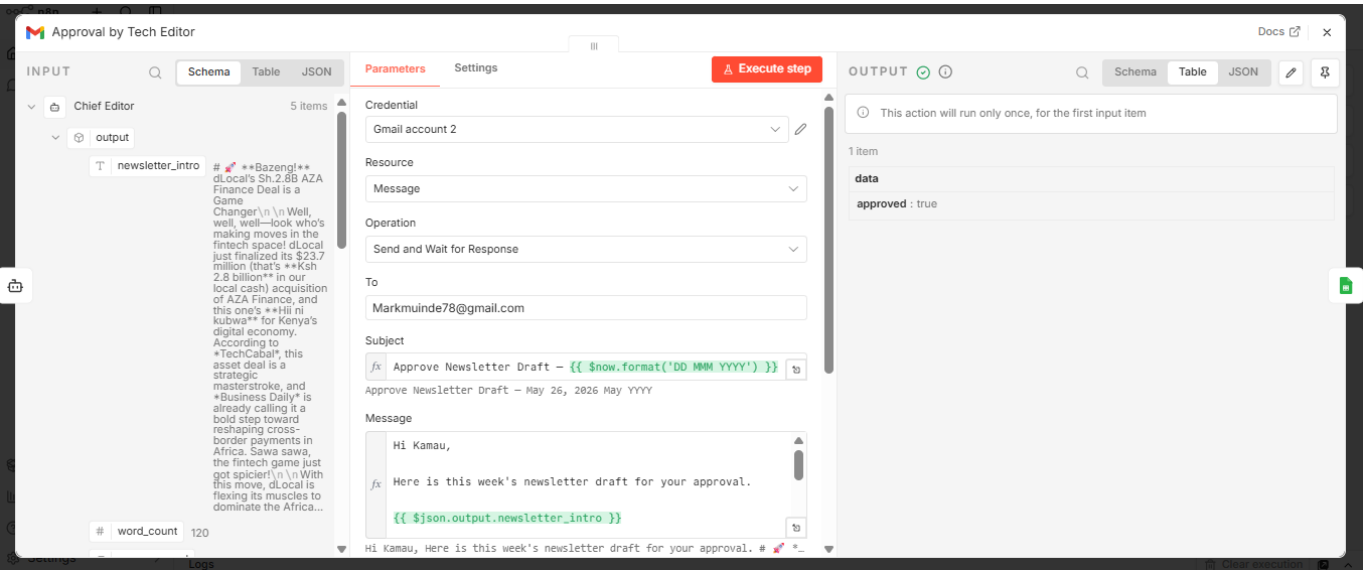
### Field Reporter — Summarize of the 5 http requests

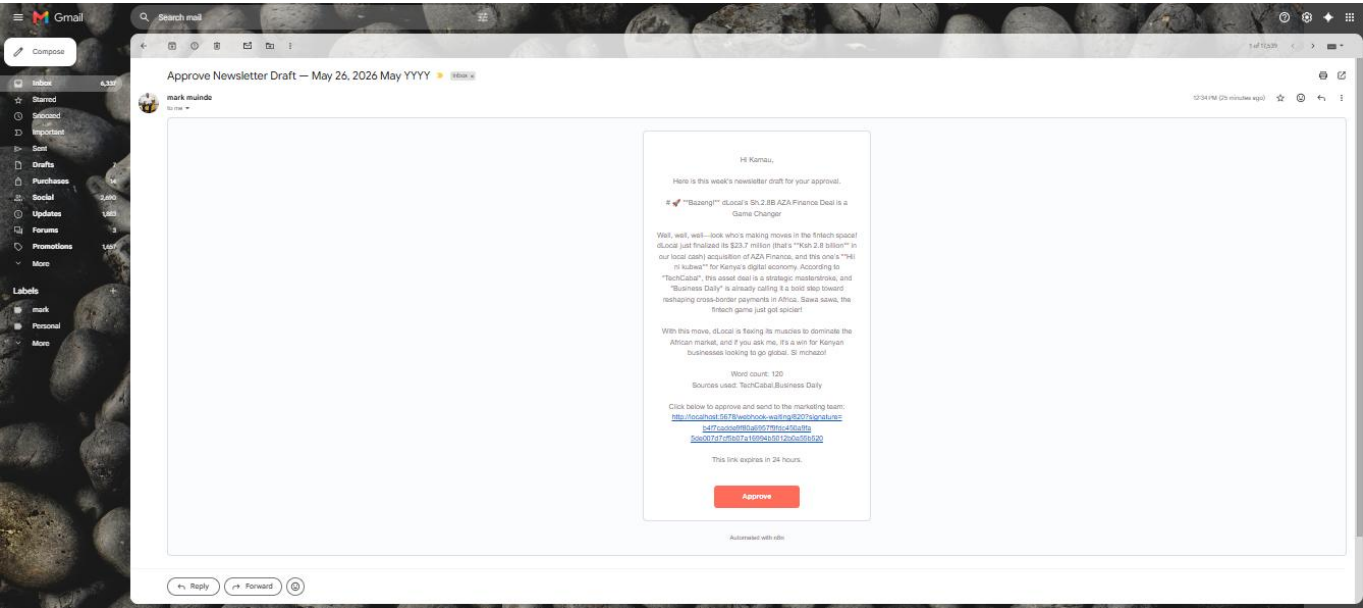


# Chief Editor — Chief Editor Mistral Large



# Approval Email — Gmail Send to Kamau for Approval

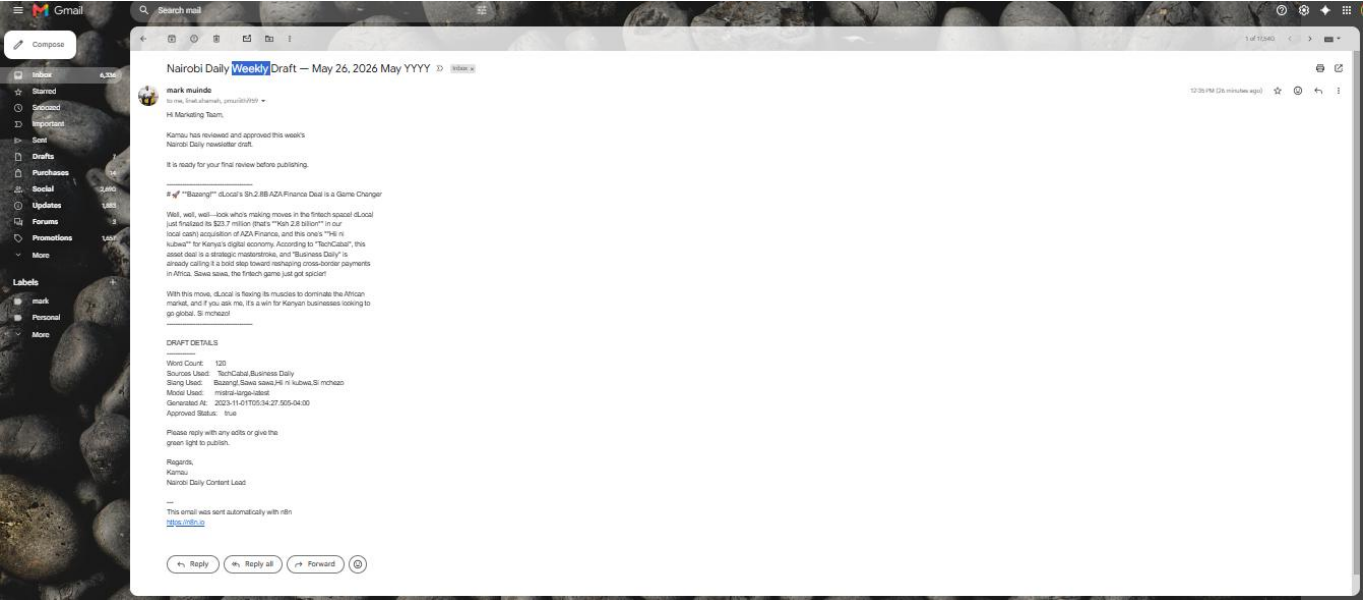




## Google Sheets Logged — Log Approved Draft

| NewsletterManager                                       |                               |                    |                                                                                                                                                                                                                                                                                                |                      |   |   |
|---------------------------------------------------------|-------------------------------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|---|---|
| File Edit View Insert Format Data Tools Extensions Help |                               |                    |                                                                                                                                                                                                                                                                                                |                      |   |   |
| 100% 123 Default... 10 B I A                            |                               |                    |                                                                                                                                                                                                                                                                                                |                      |   |   |
| E4                                                      | A                             | B                  | C                                                                                                                                                                                                                                                                                              | D                    | E | F |
| 1                                                       | Date                          | Successful Sources | Draft Body                                                                                                                                                                                                                                                                                     | Model Used           |   |   |
| 2                                                       | 2026-05-25T06:25:56.892-04:00 | 5/5 sources        | ### **VC Watu Wamechoka? Kenya Says: "Form Ni Gani?" 🇰🇪<br>Bazengi! Foreign VCs wamekuwa wakitembea huku Nairobi kama wako kwa ho<br>TechCabai and Business Daily zimeeleza how this could shake up the startup s                                                                              | mistral-large-latest |   |   |
| 3                                                       | 2026-05-25T06:42:24.124-04:00 | 5/5 sources        | ### **Bazengi! Tech Giants Flexing Muscles – Who's Next in Line?<br>South Africa's Altron just dropped a **\$30 million special dividend** after their p<br>But here's the "Form ni gani?" question: When will our local tech darlings start p<br>### **MTN's Move: Hi Ni Kubwa, Sawa Sawa? 🇰🇪 | mistral-large-latest |   |   |
| 4                                                       | 2026-05-25T11:23:31.212-04:00 | 5/5 sources        | Bazengi! The tech and telecom streets are buzzing, and this time, it's not just an<br>According to **Business Daily**, this isn't just another corporate shuffle, it's a pc<br># 🇰🇪 **Bazengi!** dLocal's Sh2.88 AZA Finance Deal is a Game Changer                                            | mistral-large-latest |   |   |
| 5                                                       | 2026-05-26T05:35:24.866-04:00 | 5/5 sources        | Well, well, well—look who's making moves in the fintech spacel dLocal just final<br>With this move, dLocal is flexing its muscles to dominate the African market, an                                                                                                                           | mistral-large-latest |   |   |
| 6                                                       |                               |                    |                                                                                                                                                                                                                                                                                                |                      |   |   |

# Email Sent to Marketing Team — Send to Marketing Team Gmail Node



## 5. Final Conclusion

This documentation now directly responds to the capstone requirements. It explains the five-source parallel workflow, individual HTTP-to-AI branches, Continue On Fail error handling, Merge/Reduce phase, Chief Editor AI constraints, backup AI fallback route, Google Sheets logging, and Gmail delivery. The screenshot slots are included so the final submission can be completed by pasting the required evidence from n8n, Gmail, and Google Sheets.